

TRANSITPULSE: LIVE BUS TRACKING AND FLEET MANAGEMENT FOR SMALL CITIES

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled “**TransitPulse: Live Bus Tracking and Fleet Management for Small Cities**” is the bonafide work of “**Aanshik Mangla (25BCE10629), Akhilesh Sreenivasan (25BCE10644), Nachiket Gadge (25BCE10852), Sourya Mishra (25BCE11063), Chinmay Sharma (25BCE10853)**” who carried out the project work under my supervision.

Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
ETA	Estimated Time of Arrival
GPS	Global Positioning System
HTML	HyperText Markup Language
ID	Identifier
ISBT	Inter State Bus Terminal
km/h	Kilometres per hour
PIN	Personal Identification Number
QR	Quick Response (code)
SMS	Short Message Service
SQLite	Lightweight file-based SQL database engine
UI	User Interface
URL	Uniform Resource Locator

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ABSTRACT

Purpose: Riders in small cities and tier-2 towns usually cannot tell when the next bus will arrive, so they wait for an unknown time or switch to private vehicles, while transport authorities have no live view of their fleet. The purpose of this project is to give commuters live bus positions, arrival times and free-seat information on any phone, and to give authorities practical tools to manage routes, stops, passes and drivers, in a form that stays usable on low-bandwidth connections.

Methodology: TransitPulse is a browser-based platform with three connected parts. A Driver Broadcaster sends each bus's position, speed and seat count from the driver's phone. A backend service stores the data in SQLite, and an ETA engine compares each bus's position and recent speed with the remaining stops on its route. A commuter tracker with a live map, and a Dispatch Center for authorities behind a role-based Civic Portal, read the same data. It was built for Patiala City Transport.

Findings: The running system shows live data for Route 101 with ETA, distance remaining, next stop, trip milestones and free seats. The Dispatch Center manages five corridors and eight geofenced stops, shows breakdown reports from drivers, logs delays between +1 and +25 minutes with their causes, and keeps a registry of pass bookings and user accounts. ETA accuracy depends on GPS quality and

traffic modelling, so displayed figures are estimates; ML-based ETA prediction, regional languages and SMS alerts are identified as future work.

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CHAPTER-1

PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

TransitPulse: Live Bus Tracking and Fleet Management for Small Cities is a live bus tracker, driver app and dispatch center for Patiala City Transport. Category: Software | Theme: Transportation & Logistics | October 4, 2026.

Bus riders in small cities and tier-2 towns usually have no way to know when the next bus will arrive. We designed TransitPulse to fix that with three connected parts: a commuter tracker that shows live bus positions, arrival times and free seats; a Driver Broadcaster that sends each bus's location, speed and seat count from the driver's phone; and a Dispatch Center where authorities manage routes, stops, passes, accounts and delay logs. We built it for Patiala City Transport. It runs in an ordinary mobile browser, needs no app installation, and is designed to stay usable on low-bandwidth connections.

1.2 Motivation for the work

- **Commuter experience:** Without arrival information, riders either arrive too early and wait, or miss the bus entirely.
- **Overcrowding:** Unpredictable schedules cause buses to bunch up, leaving some overloaded and others nearly empty.
- **Sustainability:** Every rider who switches to a private vehicle adds to congestion and emissions. Reliable information is one of the cheapest ways to keep people on public transport.
- **Administration:** Transport authorities cannot improve what they cannot see. Without location and speed data, route planning relies on guesswork.

1.3 About the project, including techniques

TransitPulse is a responsive web platform that can be opened from any phone browser, with no app installation required. It combines a commuter-facing tracker with an operator-facing fleet map and a secure Civic Portal.

The techniques used in the project are GPS telemetry sent from the bus (a low-cost GPS tracker or the driver's smartphone), an interactive Leaflet map, geofence-based recognition of bus stops, an ETA engine that compares each bus's position and recent speed with the remaining stops on its route, role-based sign-in checked against a SQLite database, and a low-bandwidth design for small-town networks. The system is described in detail in Chapters 3 to 5.

1.4 Problem Statement

In many small cities, buses run without any real-time information for riders. Schedules are unpredictable, so people either wait at stops for an unknown time or give up and use a private vehicle.

This hurts everyone involved. Commuters lose time and face overcrowded buses, authorities have no live view of their fleet, and falling ridership puts more vehicles on the road, which worsens traffic and pollution. Data cited from the Urban Mobility India Report 2024 [1] suggests that over 60% of commuters in small cities face delays because of this lack of real-time information.

Stakeholders: commuters, local transport authorities and municipal corporations.

TransitPulse was built as our own solution to the small-city transit information gap. It is submitted under hackathon problem statement and the table below shows how it meets that statement's expected outcomes.

1.5 Objective of the work

What we set out to achieve:

- Live GPS-based tracking of every bus on a web platform that works on any phone.
- Clear arrival times and route information at every stop.
- Low-bandwidth operation so the service stays usable in smaller towns.
- Practical tools for authorities to manage routes, stops, passes and drivers.

1.6 Organization of the project

The report is organized into seven chapters. Chapter 1 describes the project and its objectives. Chapter 2 investigates the related work and the current situation in small-city bus services. Chapter 3 lists the requirement artifacts. Chapter 4 presents the design methodology, architecture and user interface designs. Chapter 5 covers the technical implementation, the

working screens and the testing status. Chapter 6 discusses the project outcome and its applicability, and Chapter 7 gives the conclusions, limitations and future enhancements.

1.7 Summary

This chapter introduced TransitPulse, the lack of real-time information that riders in small cities face, the reasons the problem matters, and the objectives set for the project. The next chapter looks at related work and the existing situation.

CHAPTER-2

RELATED WORK INVESTIGATION

2.1 Introduction

This chapter describes the core area of the project and the existing situation in small-city public transport that TransitPulse responds to.

2.2 Core area of the project

The core area of the project is real-time public transport tracking for small cities. It covers GPS-based tracking of buses, estimated arrival times and route information for riders, low-bandwidth operation, and accessibility in small towns, together with fleet-management tools for transport authorities.

2.3 Existing Approaches/Methods

2.3.1 Approach/Method 1: Current practice in small cities

In many small cities, buses run without any real-time information for riders. Schedules are unpredictable, so people either wait at stops for an unknown time or give up and use a private vehicle. Authorities have no live view of their fleet, and without location and speed data, route planning relies on guesswork.

2.4 Pros and cons of the stated Approaches/Methods

Current practice (Approach 1): it needs no technology or investment, but it gives riders no arrival information, gives authorities no data on fleet location or speed, and leads to waiting, missed buses, overcrowding and falling ridership.

2.5 Issues/observations from investigation

- Over 60% of commuters in small cities face delays because of the lack of real-time information, according to data cited from the Urban Mobility India Report 2024 [1].
- Unpredictable schedules cause buses to bunch up, leaving some overloaded and others nearly empty.
- Falling ridership puts more vehicles on the road, which worsens traffic and pollution.

- Authorities cannot improve routes and timetables without location and speed data.

2.6 Summary

This chapter described the core area of the project, the current practice in small cities and the main issues observed. These issues form the basis of the requirements in the next chapter.

CHAPTER-3

REQUIREMENT ARTIFACTS

3.1 Introduction

This chapter lists the hardware and software requirements of TransitPulse and the specific requirements for its data, functions, performance and security, and look and feel.

3.2 Hardware and Software requirements

Table 1: Hardware and software requirements

Item	Requirement
Commuter device	Any phone or computer with a web browser; no app installation required
Bus-side device	A low-cost GPS tracker, or the driver's smartphone running the Driver Broadcaster screen, sending position, speed and timestamp at fixed intervals
Control-room device	A phone or computer with a web browser to open the full-screen live map and the Dispatch Center
Server	A host for the backend service (the deployed system runs on a free hosting tier at onrender.com).
Client	Mobile web browser; interactive Leaflet map
Pages	commuter_3.html, live_map.html, passes_3.html and the Civic Portal.
Backend	Backend service for ingestion, ETA calculation and operator tools.
Database	SQLite database for the latest bus positions, route definitions, stops, schedules and user accounts

3.3 Specific Project requirements

3.3.1 Data requirement

- **Bus telemetry:** bus ID / vehicle registration number, coordinates, speed and timestamp; traffic condition and vehicle status; empty seats out of 16; trip duration and last ingest ping.
- **Route data:** route ID, title, assigned bus plate, default speed, ETA and distance remaining for each corridor (five corridors, Route 101 to Route 105).

- **Stop data:** stop code, name, latitude and longitude, geofence radius (35 to 60 meters) and the routes that serve the stop (eight stops).
- **Schedule data:** timed trip milestones for each stop on a route.
- **User data:** database ID, username, PIN or password, and system role (Admin, Driver or Commuter).
- **Pass data:** booking ID, commuter username, pass scheme, tariff, category, validity, date and time issued, and status.
- **Delay-log data:** time, route, bus plate, scheduled time, actual time, deviation in minutes and main cause of delay.

3.3.2 Functions requirement

Commuter view. The commuter page lets a rider pick a bus corridor (route) and see the live status of buses on it:

- Route (corridor) selector with tabs for Route 101 to Route 105, plus an “All Buses” option.
- Live Sync indicator showing that data is being refreshed.
- Active buses on the corridor, with registration number, speed and free seats.
- Live trip status: speed in km/h, ETA in minutes, distance remaining in km and the next stop.
- Trip milestones, a stop-by-stop progress view of the journey.
- Deep links by route, for example `commuter_3.html?route=route-101`, so a route can be shared or placed on a QR code at a bus stop.

Operator and authority view.

- Full-screen live map (`live_map.html`) showing all buses on the network, useful for control-room monitoring.
- Fleet GPS telemetry that feeds the same data source as the commuter view, so riders and operators see a consistent picture.
- Civic Portal with role-based sign-in for the Department of Transport and Patiala Municipal Transport (see Section 4.5.2).

Additional service.

- Bus passes page (passes_3.html) where available passes can be viewed, extending the platform from information into service delivery.

Driver Broadcaster. The driver selects the assigned corridor and direction, enters the vehicle registration number, reports the traffic condition and vehicle status, updates the live seat map and starts the trip to broadcast position, speed and seat availability (Section 4.5.3).

Dispatch Center. Authorities monitor the live fleet map, review analytics and delay logs, view the pass bookings registry, create or delete routes, register bus stops with GPS geofences, and manage user accounts (Section 5.3).

3.3.3 Performance and security requirement

Many small-town riders have limited mobile data, so we designed TransitPulse for low bandwidth. The following measures are recommended:

- Send only compact data (bus ID, coordinates, speed, time) rather than full page reloads.
- Refresh at moderate intervals and back off when the connection is weak.
- Keep pages light with minimal images and scripts, and cache static assets such as route and stop data.
- Show the last known position with a “last updated” time when the connection drops, instead of an empty screen.
- Offer a text-only or SMS fallback for riders without smartphones (future work).

For security, access to the Civic Portal is checked on the server side against a SQLite users database rather than hard-coded in the page, and the system uses role-based access for commuters, drivers and administrators. Passwords are hidden on screen, but for a production deployment they should also be stored as salted hashes rather than plain text.

3.3.4 Look and Feel requirements

TransitPulse is a responsive web platform that opens from any phone browser with no app installation. The commuter page shows a green Live Sync badge, route tabs with the number of live buses, an interactive map with zoom controls that can be expanded to full screen, a stop-by-stop milestone timeline and a progress bar. The driver’s seat map shows empty seats in green and occupied seats in red, with a large green Start Trip button and a red Standby badge before the trip starts (see Section 4.5).

3.3.5 Accessibility requirements

The service must run in any mobile browser so that it is accessible in small towns. Route-specific links (for example `commuter_3.html?route=route-101`) can be printed as QR codes at bus stops, and a text-only or SMS fallback for riders without smartphones is planned as future work.

3.4 Summary

This chapter listed the hardware and software requirements and the data, functional, performance, security, look-and-feel and accessibility requirements of TransitPulse.

CHAPTER-4

DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and goal

TransitPulse is designed as three connected parts that share one data source: a commuter tracker that shows live bus positions, arrival times and free seats; a Driver Broadcaster that sends each bus's location, speed and seat count from the driver's phone; and a Dispatch Center where authorities manage routes, stops, passes, accounts and delay logs. Because riders and operators read the same data, both see a consistent picture.

The goal of the design is to give riders reliable real-time information through a lightweight, accessible web platform, to give drivers a simple way to broadcast their trips, and to give authorities a secure place to manage the network, while staying usable on low-bandwidth connections. The novelty of the design lies in combining all three parts in one browser-based, no-installation platform built for small cities.

4.2 Functional modules design and analysis

- **Commuter tracker (commuter_3.html):** route selector, live map, active buses, trip status and trip milestones.
- **Full-screen live map (live_map.html):** all buses on the network for control-room monitoring.
- **Bus passes page (passes_3.html):** view and book available passes.
- **Driver Broadcaster:** trip setup, live readings, seat map and Start Trip.
- **Civic Portal:** sign-in and account creation with role-based access for commuters, drivers and administrators.
- **Dispatch Center (Admin Command Room):** Live Fleet GPS Map, Analytics & Logs, Pass Bookings Registry, Routes & Sequences, Bus Stops & GPS and User Credentials Directory.

The modules share the same backend and database, so a value entered in one module (for example a driver reporting a flat tyre or a seat count) is visible in the others.

4.3 Software Architectural designs

The platform follows a standard telemetry pipeline:

1. **Data capture:** A low-cost GPS tracker or the driver's smartphone running the Driver Broadcaster screen on each bus sends position, speed and timestamp at fixed intervals.
2. **Ingestion and storage:** A backend service receives the updates and stores the latest position of every bus, along with route definitions, stops, schedules and user accounts in a SQLite database.
3. **ETA engine:** The backend compares each bus's position and recent speed with the remaining stops on its route to compute distance remaining and estimated arrival time.
4. **Delivery:** The web front end polls or subscribes to the backend and refreshes the map, bus card and milestones.
5. **Operator tools:** The full-screen map, passes page and Civic Portal use the same backend.

4.4 Subsystem services

- **Ingestion and storage service:** receives bus updates and keeps the latest position of every bus in SQLite.
- **ETA service:** computes distance remaining, next stop and estimated arrival time from position and recent speed.
- **Geofence and milestone service:** each stop has a geofence radius, which lets the system recognize when a bus has reached a stop and drives the milestone updates seen by commuters.
- **Authentication service:** the Civic Portal verifies credentials against a SQLite users database on the server side and applies role-based access.
- **Delay-log and pass services:** record scheduled versus actual times with the cause of delay, and record pass bookings, connecting ticketing revenue to the same platform that tracks the buses.

4.5 User Interface designs

The screenshots below come from the running TransitPulse system. Figure 1 shows the commuter tracker, and Figures 2 and 3 show the sign-in and registration screens of the Civic Portal. Figures 4 to 6 show the Driver Broadcaster screen and the driver sign-up.

4.5.1 Commuter tracker

Figure 1: Live commuter tracker (Route 101: Bus Stand ⇌ Railway Station)

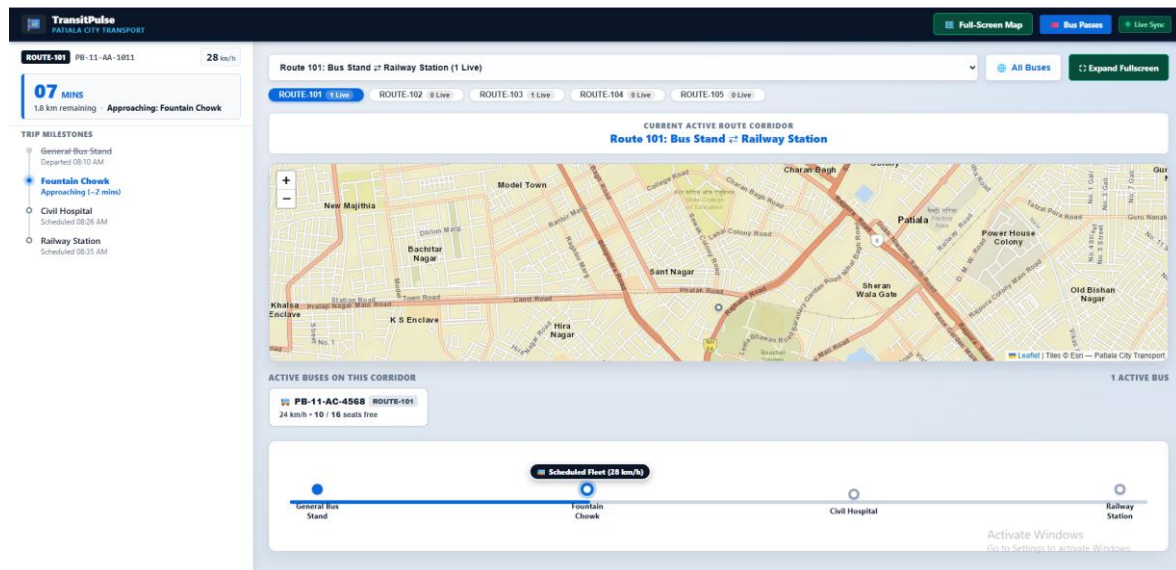


Figure 1: TransitPulse live commuter tracker for Route 101

This is the main screen a commuter uses. Its parts are:

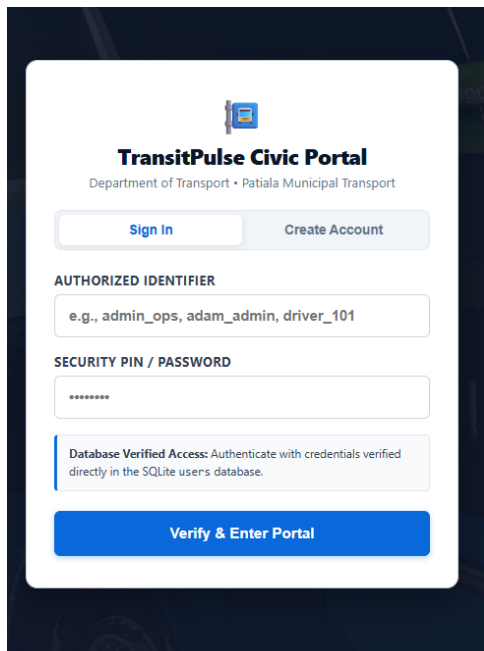
- **Top bar:** The TransitPulse and Patiala City Transport branding, with shortcuts to the Full-Screen Map and Bus Passes, and a green Live Sync badge that shows data is updating.
- **Route selector:** A dropdown plus one tab per corridor (Route 101 to Route 105). Each tab shows how many buses are live; here Route 101 has 1 live bus and the others have 0.
- **Live map:** An interactive Leaflet map of Patiala with zoom controls. It sits under the corridor title and can be expanded to full screen.
- **Active buses card:** Lists each bus on the corridor with its registration number, route, speed in km/h and free seats. In this capture, bus PB-11-AC-4568 is moving at 24 km/h with 10 of 16 seats free, which helps riders judge crowding before boarding.
- **Left panel (trip status):** Shows the route's scheduled bus (PB-11-AA-1011) with its default speed (28 km/h), a large ETA in minutes, distance remaining (1.8 km) and the

next stop (Fountain Chowk). These defaults come from the route table that operators manage (Figure 10), which is why they differ slightly from the live bus card.

- **Trip milestones:** A stop-by-stop timeline. General Bus Stand is marked departed at 08:10 AM, Fountain Chowk is marked approaching in about 2 minutes, and Civil Hospital (08:26 AM) and Railway Station (08:35 AM) are scheduled.
- **Progress bar:** A horizontal line from General Bus Stand to Railway Station with a marker showing where the bus currently sits between stops.

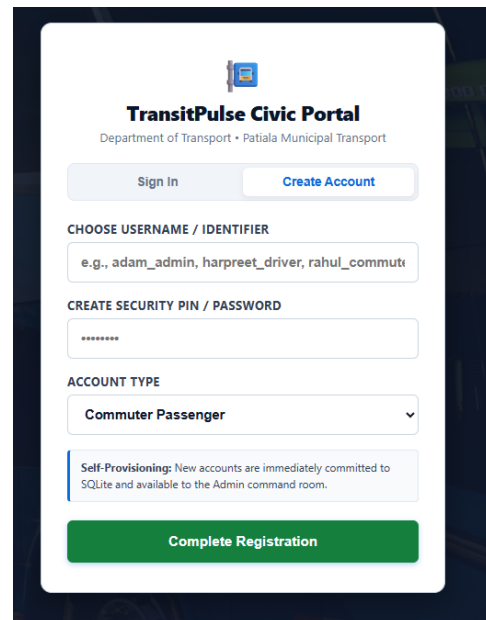
4.5.2 Civic Portal sign-in and account creation

Figures 2 and 3: Civic Portal sign-in and account creation



The screenshot shows the 'TransitPulse Civic Portal' sign-in interface. At the top, it says 'Department of Transport • Patiala Municipal Transport'. Below this are two buttons: 'Sign In' and 'Create Account'. The 'Sign In' section has two input fields: 'AUTHORIZED IDENTIFIER' with a hint 'e.g., admin_ops, adam_admin, driver_101' and 'SECURITY PIN / PASSWORD' with a hint '*****'. A blue box below these fields states 'Database Verified Access: Authenticate with credentials verified directly in the SQLite users database.' At the bottom is a large blue button labeled 'Verify & Enter Portal'.

Figure 2: Civic Portal sign-in



The screenshot shows the 'TransitPulse Civic Portal' account creation interface. At the top, it says 'Department of Transport • Patiala Municipal Transport'. Below this are two buttons: 'Sign In' and 'Create Account'. The 'Create Account' section has three input fields: 'CHOOSE USERNAME / IDENTIFIER' with a hint 'e.g., adam_admin, harpreet_driver, rahul_commuter', 'CREATE SECURITY PIN / PASSWORD' with a hint '*****', and 'ACCOUNT TYPE' with a dropdown menu showing 'Commuter Passenger'. A blue box below these fields states 'Self-Provisioning: New accounts are immediately committed to SQLite and available to the Admin command room.' At the bottom is a large green button labeled 'Complete Registration'.

Figure 3: Civic Portal account creation

Sign-in (Figure 2). The Civic Portal is the secure entry point for the Department of Transport and Patiala Municipal Transport. Users enter an authorized identifier (for example an admin, operator or driver ID) and a security PIN or password. The on-screen note states that credentials are verified against a SQLite users database, which means access is checked on the server side rather than hard-coded in the page.

Account creation (Figure 3). New users can self-register by choosing a username, a PIN or password, and an account type. The account type dropdown (Commuter Passenger is shown) points to role-based access, with different experiences for commuters, drivers and administrators. The on-screen note says new accounts are saved immediately to SQLite and

become visible in the Admin command room, so authorities can see and manage who is using the system.

Table 2: What the screenshots show about the solution

Capability	Evidence in the screenshots
Real-time bus tracking	Live map, Live Sync badge and live-bus counts per route (Figure 1)
Estimated arrival time and route information	ETA in minutes, distance remaining, next stop and timed milestones (Figure 1)
Commuter convenience	Free-seat count per bus and one-tap switching between five routes (Figure 1)
Support for authorities	Role-based login and an Admin command room that receives new accounts (Figures 2 and 3)
Low-cost, small-city fit	Browser-based interface and a lightweight SQLite database (Figures 1 to 3)

4.5.3 Driver Broadcaster (Onboard Driver App)

The live data that commuters see comes from the bus itself. Each driver uses a dedicated Driver Broadcaster screen that sends the bus’s position, speed and seat availability to the platform. Figures 4 to 6 show this screen and the driver sign-up.

Figure 4: Driver Broadcaster, trip setup and live readings

Figure 4: TransitPulse Driver Broadcaster trip setup screen

Before starting a trip, the driver fills in a short form. The screen contains:

- **Status badge:** A red “Standby” badge in the top right shows the bus is not yet broadcasting.

- **Assigned corridor and direction:** A dropdown to choose the route the bus is serving, so the bus appears on the correct corridor in the commuter tracker.
- **Vehicle registration number:** A text field (for example PB-11-AA-1011) that identifies the bus.
- **Traffic condition and vehicle status:** Dropdowns, set to “Normal” and “Running Fine” here, which let the driver report congestion or a breakdown for use in ETA and alerts.
- **Live readings:** Speed (0.0 km/h before the trip), empty seats (16 / 16), trip duration (00:00), last ingest ping and live latitude and longitude. These fill in once the trip starts and show the driver that data is reaching the server.

Figure 5: Live seat availability and Start Trip

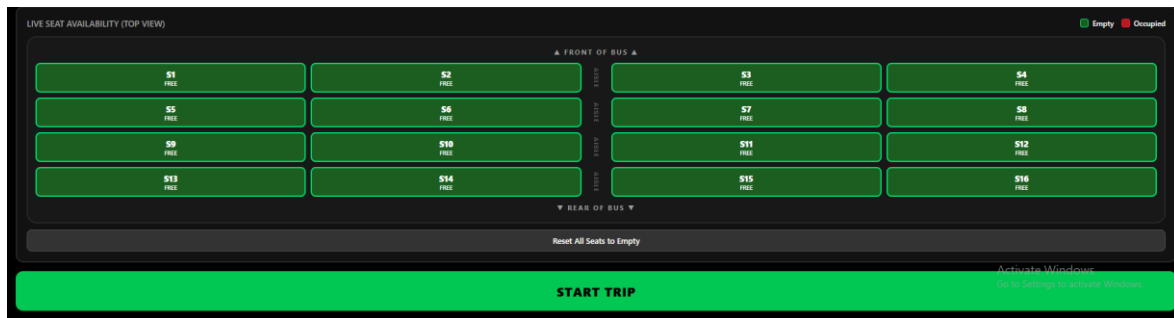


Figure 5: TransitPulse Driver Broadcaster seat map and Start Trip button

Below the form is a top-view seat map of the 16-seat bus (S1 to S16), laid out in four rows with an aisle between the left and right pairs and the front and rear of the bus labeled. Seats are shown green when empty and red when occupied, and the driver can reset them all with one button. This count is what commuters see as “seats free” on the tracker in Figure 1. A large green **Start Trip** button begins broadcasting.

Figure 6: Driver account registration

The screenshot shows the 'TransitPulse Civic Portal' registration interface. At the top, it says 'Department of Transport • Patiala Municipal Transport'. There are 'Sign In' and 'Create Account' buttons. Below, the 'CHOOSE USERNAME / IDENTIFIER' field has a hint: 'e.g., adam_admin, harpreet_driver, rahul_commuti'. The 'CREATE SECURITY PIN / PASSWORD' field is masked with dots. The 'ACCOUNT TYPE' dropdown is set to 'Onboard Bus Driver'. A note states: 'Self-Provisioning: New accounts are immediately committed to SQLite and available to the Admin command room.' At the bottom is a green 'Complete Registration' button.

Figure 6: Civic Portal registration with the Onboard Bus Driver account type

Drivers get access through the same Civic Portal as other users. On the Create Account tab, the driver selects the account type **Onboard Bus Driver** instead of Commuter Passenger. The account is saved to the SQLite database and appears in the Admin command room, where authorities can see which drivers are registered.

Table 3: What the driver screens add to the solution

Capability	Evidence in the screenshots
Source of real-time GPS data	Live coordinates, speed and last ingest ping on the driver screen (Figure 4)
Route assignment	Assigned corridor and direction dropdown plus vehicle registration number (Figure 4)
Crowding information for commuters	16-seat live seat map and empty-seat counter (Figures 4 and 5)
Operational reporting	Traffic condition and vehicle status fields (Figure 4)
Controlled driver access	Onboard Bus Driver account type in the Civic Portal (Figure 6)

4.6 Summary

This chapter presented the methodology and goal of TransitPulse, its functional modules, the telemetry pipeline architecture, the subsystem services and the user interface designs of the commuter tracker, the Civic Portal and the Driver Broadcaster.

CHAPTER-5

TECHNICAL IMPLEMENTATION & ANALYSIS

5.1 Outline

This chapter describes how the design was implemented, the working screens of the Dispatch Center, the prototype that was deployed, and the testing and validation status.

5.2 Technical coding and code solutions

The implementation follows the pipeline described in Section 4.3. The main code solutions are:

- A Driver Broadcaster page that sends the bus's position, speed, timestamp, seat count, traffic condition and vehicle status to the backend.
- A backend service that stores the latest position of every bus, route definitions, stops, schedules, delay logs, pass bookings and user accounts in SQLite.
- An ETA engine that compares each bus's position and recent speed with the remaining stops on its route to compute distance remaining and estimated arrival time.
- Geofence checks (35 to 60 meter radius per stop) that detect when a bus has reached a stop and update the trip milestones.
- Front-end pages (commuter_3.html, live_map.html, passes_3.html and the Civic Portal) that refresh the Leaflet map, bus card and milestones; the Dispatch Center fleet map refreshes automatically every 3 seconds.

5.3 Working Layout of Forms

Operator Dispatch Center (Admin Command Room). Transport authorities manage the whole network from the TransitPulse Patiala Dispatch Center, which is shown to a signed-in Central Dispatcher at the Patiala Municipal Depot. A left-hand menu groups six tools under four headings: Operations (Live Fleet GPS Map, Analytics & Logs), Fares & Concessions (Pass Bookings Registry), Network Infrastructure (Routes & Sequences, Bus Stops & GPS) and System Security (User Credentials Directory). Figures 7 to 12 show each tool. The sign-in, account creation and Driver Broadcaster forms are shown in Figures 2 to 6.

Figure 7: Live Fleet GPS Tracking Room

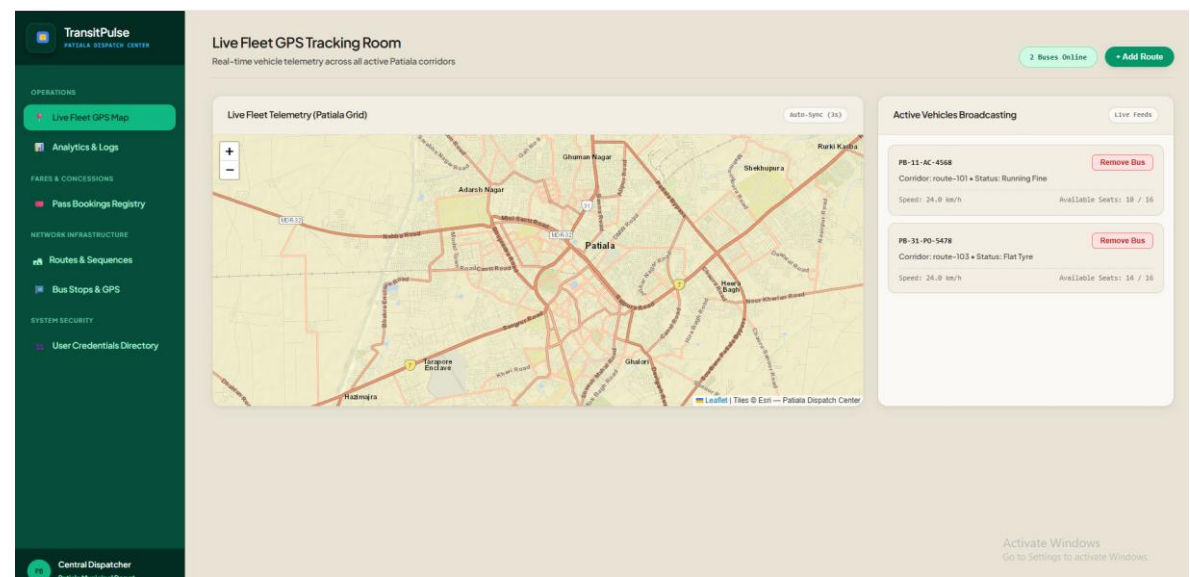


Figure 7: TransitPulse operator live fleet GPS map

The landing screen is a live map of Patiala that refreshes automatically every 3 seconds. A badge shows how many buses are online (2 here), and the panel on the right lists each broadcasting vehicle with its plate number, corridor, status, speed and available seats. In this capture, bus PB-11-AC-4568 is on route-101 with status “Running Fine”, 24 km/h and 10 of 16 seats free, while PB-31-PO-5478 is on route-103 with status “Flat Tyre”, 24 km/h and 14 of 16 seats free. These values are the ones the drivers enter in the Driver Broadcaster (Section 4.5.3), so the dispatcher sees breakdowns as they are reported. Buttons allow a bus to be removed from the feed or a new route to be added.

Figure 8: Operational Analytics & Delay Logs

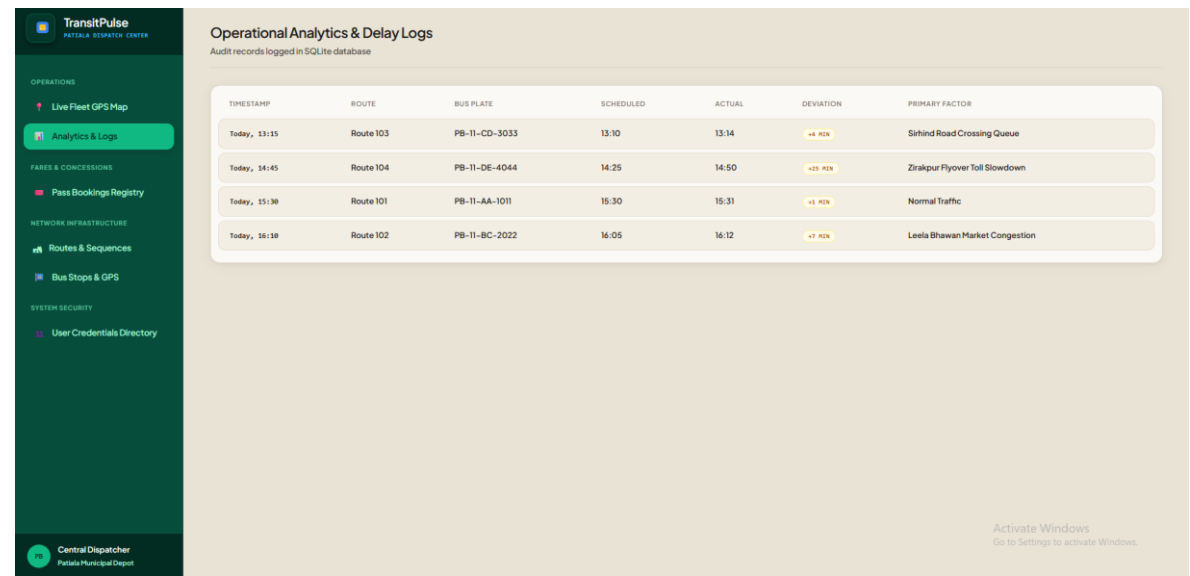
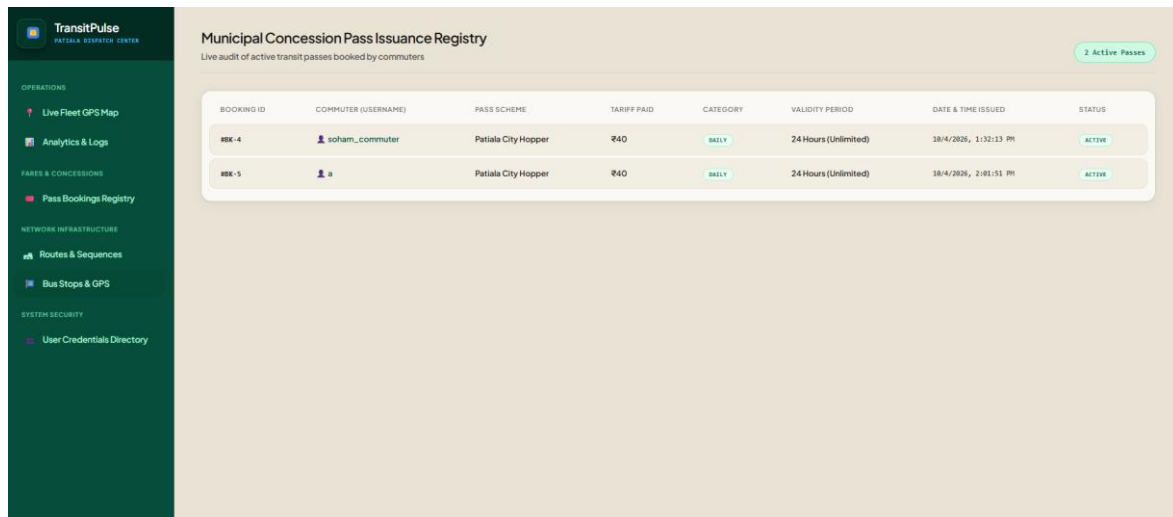


Figure 8: TransitPulse operational analytics and delay logs

This screen is an audit log stored in the SQLite database. Each row records the time, route, bus plate, scheduled time, actual time, the deviation in minutes and the main cause of delay. The records shown range from +1 minute on Route 101 (normal traffic) to +25 minutes on Route 104 (Zirakpur flyover toll slowdown), with other causes such as a road-crossing queue on Route 103 and market congestion on Route 102. Authorities can use these logs to find chronic bottlenecks and adjust timetables.

Figure 9: Municipal Concession Pass Issuance Registry

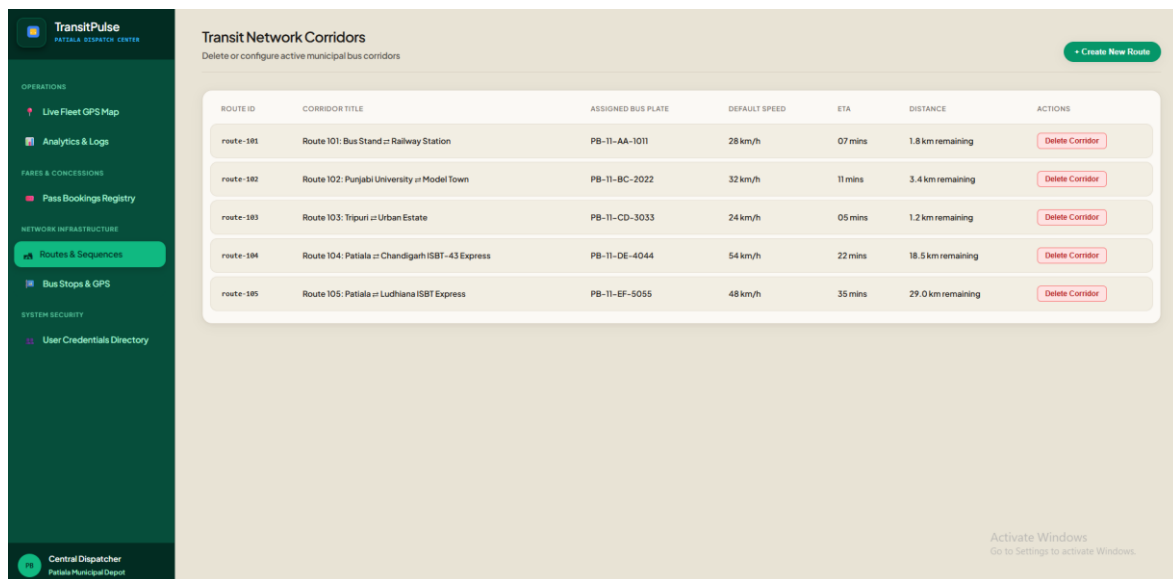


BOOKING ID	COMMUTER (USERNAME)	PASS SCHEME	TARIFF PAID	CATEGORY	VALIDITY PERIOD	DATE & TIME ISSUED	STATUS
HBK-4	soham_commuter	Patiala City Hopper	₹40	DAILY	24 Hours (Unlimited)	18/4/2026, 1:32:13 PM	ACTIVE
HBK-5	a	Patiala City Hopper	₹40	DAILY	24 Hours (Unlimited)	18/4/2026, 2:01:51 PM	ACTIVE

Figure 9: TransitPulse pass bookings registry

This registry gives a live audit of the transit passes booked by commuters through the passes page. Each entry shows the booking ID, the commuter's username, the pass scheme (Patiala City Hopper), the tariff paid (₹40), the category (Daily), the validity (24 hours, unlimited), the date and time issued, and the status (Active). A counter at the top shows the number of active passes (2). This connects ticketing revenue to the same platform that tracks the buses.

Figure 10: Transit Network Corridors

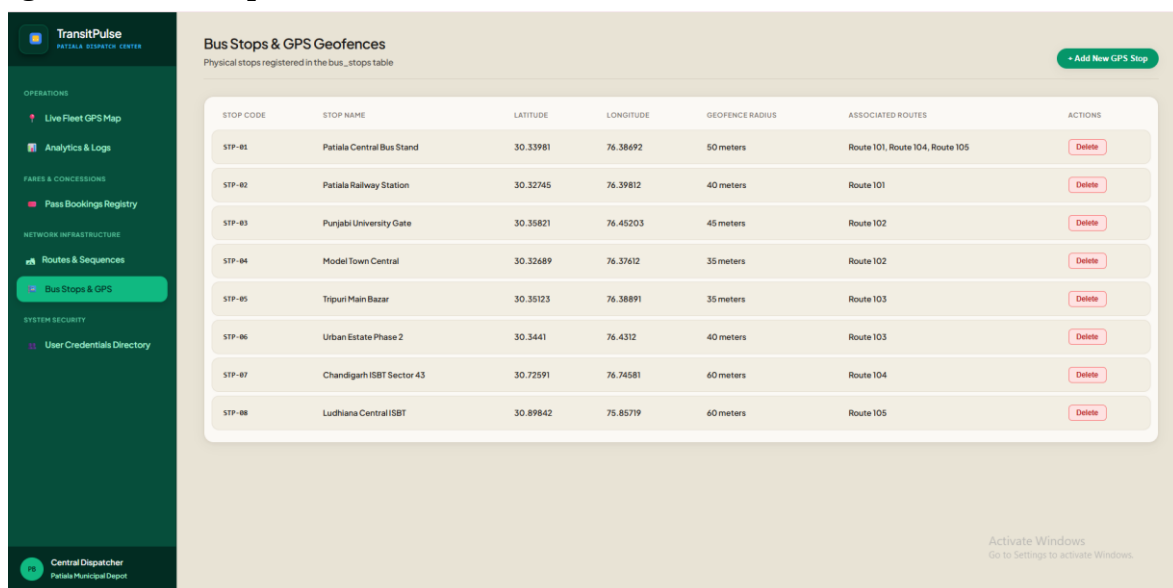


ROUTE ID	CORRIDOR TITLE	ASSIGNED BUS PLATE	DEFAULT SPEED	ETA	DISTANCE	ACTIONS
route-101	Route 101: Bus Stand to Railway Station	PB-11-AA-1011	28 km/h	07 mins	1.8 km remaining	Delete Corridor
route-102	Route 102: Punjabi University to Model Town	PB-11-BC-2022	32 km/h	11 mins	3.4 km remaining	Delete Corridor
route-103	Route 103: Tripuri to Urban Estate	PB-11-CD-3033	24 km/h	05 mins	1.2 km remaining	Delete Corridor
route-104	Route 104: Patiala to Chandigarh ISBT-43 Express	PB-11-DE-4044	54 km/h	22 mins	18.5 km remaining	Delete Corridor
route-105	Route 105: Patiala to Ludhiana ISBT Express	PB-11-EF-5055	48 km/h	35 mins	29.0 km remaining	Delete Corridor

Figure 10: TransitPulse transit network corridors

This table defines the routes that appear in the commuter tracker. Each corridor has a route ID, a title, an assigned bus plate, a default speed, an ETA and a distance remaining. There are five corridors: Route 101 (Bus Stand to Railway Station), Route 102 (Punjabi University to Model Town), Route 103 (Tripuri to Urban Estate), Route 104 (Patiala to Chandigarh ISBT-43 Express) and Route 105 (Patiala to Ludhiana ISBT Express). The default values, such as 28 km/h and 07 minutes for Route 101, are the scheduled figures shown on the left panel of the commuter tracker in Figure 1. Operators can create a new route or delete an existing corridor.

Figure 11: Bus Stops & GPS Geofences



STOP CODE	STOP NAME	LATITUDE	LONGITUDE	GEOFENCE RADIUS	ASSOCIATED ROUTES	ACTIONS
STP-01	Patiala Central Bus Stand	30.33981	76.38692	50 meters	Route 101, Route 104, Route 105	Delete
STP-02	Patiala Railway Station	30.32745	76.39812	40 meters	Route 101	Delete
STP-03	Punjabi University Gate	30.35821	76.45203	45 meters	Route 102	Delete
STP-04	Model Town Central	30.32689	76.37612	35 meters	Route 102	Delete
STP-05	Tripuri Main Bazar	30.35123	76.38891	35 meters	Route 103	Delete
STP-06	Urban Estate Phase 2	30.3441	76.4312	40 meters	Route 103	Delete
STP-07	Chandigarh ISBT Sector 43	30.72591	76.74581	60 meters	Route 104	Delete
STP-08	Ludhiana Central ISBT	30.89842	75.85719	60 meters	Route 105	Delete

Figure 11: TransitPulse bus stops and GPS geofences

Eight physical stops are registered in the bus stops table, each with a stop code, name, latitude and longitude, a geofence radius (35 to 60 meters) and the routes that serve it. For example, Patiala Central Bus Stand (STP-01) is served by Routes 101, 104 and 105 with a 50-meter geofence. The geofence radius lets the system recognize when a bus has reached a stop, which is what drives the milestone updates seen by commuters. New stops can be added with their GPS position.

Figure 12: Authorized User Accounts & Credentials Directory

DATABASE ID	USERNAME IDENTIFIER	SECURITY PIN / PASSWORD	SYSTEM ROLE	ACTIONS
#3	admin_ops	*****	ADMIN	Delete
#12	aman_driver	*****	DRIVER	Delete
#13	rohan_driver	*****	DRIVER	Delete
#14	soham_commuter	*****	COMMUTER	Delete
#15	rahuL_commuter	*****	COMMUTER	Delete
#16	MANISH_DRIVER	*****	DRIVER	Delete
#17	adam_admin	*****	ADMIN	Delete
#18	a	*****	COMMUTER	Delete
#19	ab	*****	DRIVER	Delete
#20	abc	*****	ADMIN	Delete

Figure 12: TransitPulse user accounts and credentials directory

This directory lists every account in the system with its database ID, username, masked PIN or password and system role (Admin, Driver or Commuter). The capture shows ten accounts: three admins, four drivers and three commuters. Administrators can delete accounts or add staff accounts directly. Passwords are hidden on screen, but for a production deployment they should also be stored as salted hashes rather than plain text.

Table 4: What the operator screens add to the solution

Capability	Evidence in the screenshots
Fleet-wide live monitoring	Auto-syncing map and list of broadcasting vehicles with status and seats (Figure 7)
Incident awareness	Vehicle status such as “Flat Tyre” visible to the dispatcher (Figure 7)
Performance auditing	Scheduled versus actual times and delay causes per trip (Figure 8)
Fare and pass management	Pass issuance registry with tariff and validity (Figure 9)
Network configuration	Create or delete corridors and register stops with geofences (Figures 10 and 11)

Capability	Evidence in the screenshots
Access control	Directory of admin, driver and commuter accounts (Figure 12)

5.4 Prototype submission

- The deployed pages are reachable and show the commuter interface, the full-screen map link and the passes link.
- TransitPulse live deployment: https://transitpulse-2.onrender.com/commuter_3.html?route=route-101 [2]

5.5 Test and validation

- The screenshots in Section 4.5 show the system working with live data for Route 101. An automated text check of the page, made while the hosting service may have been waking from idle, only showed loading placeholders, so re-verify the live feed shortly before any demonstration.
- ETA figures in the demo should be treated as estimates (see Section 7.2).

5.6 Performance Analysis (Graphs/Charts)

The Dispatch Center fleet map refreshes automatically every 3 seconds (Figure 7), and the Operational Analytics & Delay Logs screen records delays from +1 minute on Route 101 to +25 minutes on Route 104 (Figure 8), which authorities can use to find chronic bottlenecks and adjust timetables. Low-bandwidth behaviour follows the measures listed in Section 3.3.3.

5.7 Summary

This chapter described the implementation of TransitPulse, the working screens of the Dispatch Center, the deployed prototype and the testing status.

CHAPTER-6

PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This chapter summarizes what the project delivered, its significant outcomes and where it can be applied.

6.2 Key implementations outlines of the System

- A commuter tracker with live map, route tabs for five corridors, active-bus cards with free seats, ETA and trip milestones.
- A Driver Broadcaster that sends location, speed and seat count from the driver's phone.
- A Dispatch Center with live fleet map, analytics and delay logs, pass bookings registry, route and stop management, and a user credentials directory.
- A Civic Portal with role-based sign-in for commuters, drivers and administrators, backed by SQLite.
- A bus passes page, extending the platform from information into service delivery.

6.3 Significant project outcomes

- A working system built for Patiala City Transport that runs in an ordinary mobile browser and needs no app installation.
- Live data shown for Route 101, with riders and operators seeing a consistent picture from the same data source.
- Five corridors and eight bus stops with 35 to 60 meter geofences configured in the network tables.
- Delay logging that ranges from +1 to +25 minutes per trip and identifies causes of delay, and a pass registry with 2 active Patiala City Hopper passes (₹40 daily).
- A user directory of ten accounts (three admins, four drivers and three commuters).

6.4 Project applicability on Real-world applications

- **For commuters:** Less waiting time, better trip planning and more confidence in public transport.

- **For transport authorities:** Visibility into fleet location and speed, supporting better scheduling and accountability.
- **For the city:** Higher bus ridership, lower private vehicle use, and reduced congestion and emissions.

With a stable data feed, a well-tested ETA engine and a low-bandwidth focus, TransitPulse can serve as a replicable model for other tier-2 towns.

6.5 Inference

TransitPulse shows that real-time bus information, driver reporting and fleet management can be delivered together through a lightweight browser-based platform suited to small cities.

CHAPTER-7

CONCLUSIONS AND RECOMMENDATION

7.1 Outline

This chapter gives the limitations of the system, the recommended future enhancements and the conclusion of the project.

7.2 Limitation/Constraints of the System

- If the backend runs on a free hosting tier, cold starts can delay the first load by up to a minute. A paid tier or a keep-alive ping is advisable for a live demo.
- ETA accuracy depends on GPS quality and on how traffic and stops are modeled. Figures in the demo should be treated as estimates.
- The live feed should be re-verified shortly before any demonstration (see Section 5.5).
- Passwords are hidden on screen, but for a production deployment they should also be stored as salted hashes rather than plain text.
- A text-only or SMS fallback for riders without smartphones is not yet available (future work).

7.3 Future Enhancements

- Machine-learning ETA prediction using historical trip data, time of day and traffic patterns.
- Crowding indicators built on the existing free-seat count.
- Regional language support (Punjabi and Hindi) and voice or SMS alerts.
- Arrival notifications for a chosen stop.
- Digital payment and pass purchase integrated with the passes page.
- Analytics dashboard for authorities showing punctuality, route usage and idle time.
- Open data API so other city services and third-party apps can use the feed.

7.4 Inference

TransitPulse gives commuters in a small city reliable, real-time information about their buses through a lightweight, accessible web platform. It gives drivers a simple way to broadcast their

trips and gives authorities a secure place to manage the network. With a stable data feed, a well-tested ETA engine and a low-bandwidth focus, it can serve as a replicable model for other tier-2 towns.

APPENDIX A

Live deployment and pages

- Commuter tracker: https://transitpulse-2.onrender.com/commuter_3.html?route=route-101
- Full-screen live map: [live_map.html](#)
- Bus passes page: [passes_3.html](#)

REFERENCES

- [1] Urban Mobility India Report 2024
- [2] TransitPulse live deployment. [Online]. Available: https://transitpulse-2.onrender.com/login_2.html